Detecting Human Activity Using Hidden Markov Models

1. Background and Motivation

Human Activity Recognition (HAR) is an important application in machine learning that enables systems to infer human behavior using sensor data. It is widely used in wearable devices, healthcare monitoring systems, smart homes, and fitness tracking applications.

In this project, a Hidden Markov Model (HMM) is used to classify physical activities using motion sensor data collected from smartphones. The sensors used include accelerometers and gyroscopes, which capture motion dynamics across three axes (x, y, z).

Unlike simple classifiers, HMMs are particularly suited for this task because they model **temporal sequences**, allowing the system to learn how activities transition over time. This makes them well-suited for activity recognition where physical movements follow natural temporal patterns.

The system focuses on recognizing four daily activities: **Still, Standing, Walking,** and **Jumping**

2. Data Collection and Preprocessing

2.1 Data Collection Setup

Motion data were recorded using smartphone sensors and stored as CSV files containing timestamped sensor readings.

Each recording contains measurements from:

- **Accelerometer (x, y, z)** – measuring linear acceleration (m/s^2)
- **Gyroscope (x, y, z)** – measuring rotational velocity (rad/s)

Each activity recording lasts roughly **10 seconds**.

The dataset was organized into multiple clips where each clip corresponds to a single activity recording.

Dataset Split

The dataset was divided into training and testing subsets.

Dataset	File IDs	Clips
Training	1–12	48 clips

Testing	13–15	12 clips

The activities used in the project are mapped to numerical labels:

Activity	Label
Still	0
Standing	1
Walking	2
Jumping	3

2.2 Preprocessing Steps

Several preprocessing steps were applied to prepare the sensor data for modeling.

1. Data Loading

Accelerometer and gyroscope data were loaded from CSV files and merged into a single dataset containing six motion signals:

- Acceleration: **`acc_x`**, **`acc_y`**, **`acc_z`**
- Gyroscope: **`gyro_x`**, **`gyro_y`**, **`gyro_z`**

The pipeline automatically detects the relevant column names and aligns timestamps between sensors.

2. Signal Normalization

All six sensor signals were standardized using **z-score normalization** so that each signal has:

- Mean = 0
- Standard deviation = 1

This step ensures that features from different sensors contribute equally to the model.

3. Sampling Rate Harmonization

Different recordings were collected at slightly different sampling rates.

The pipeline estimated sampling rates for all clips and computed the **median sampling rate**:

- Detected sampling rates: ~50 Hz and ~103 Hz
- **Target sampling rate:** ~76.85 Hz

All signals were resampled to this target rate to ensure consistency.

3. Feature Extraction

Each window was converted into a feature vector containing both **time-domain** and **frequency-domain** features.

A total of **75 features** were extracted per window.

Time-Domain Features

For each sensor axis:

- Mean
- Standard deviation
- Variance
- Root Mean Square (RMS)
- Signal Magnitude Area

Additional features included:

- Correlation between axes
- Acceleration magnitude

These features capture the intensity and variability of movements.

Frequency-Domain Features

Using the Fast Fourier Transform (FFT), the following features were extracted:

- Dominant frequency
- Spectral energy
- Spectral entropy

These features help distinguish activities with periodic motion (such as walking or jumping).

4. Dataset Construction

The dataset construction process iterates through each activity and file ID:

1. Load accelerometer and gyroscope files
2. Merge and resample signals
3. Normalize sensor values
4. Apply sliding windows
5. Extract features for each window

The final dataset consists of:

- **Feature vectors (observations)**
- **Activity labels**
- **Clip boundaries** used during HMM training

5. HMM Setup and Implementation Details

The activity recognition model was implemented using the **Gaussian Hidden Markov Model** from the `hmmlearn` library.

Model Parameters

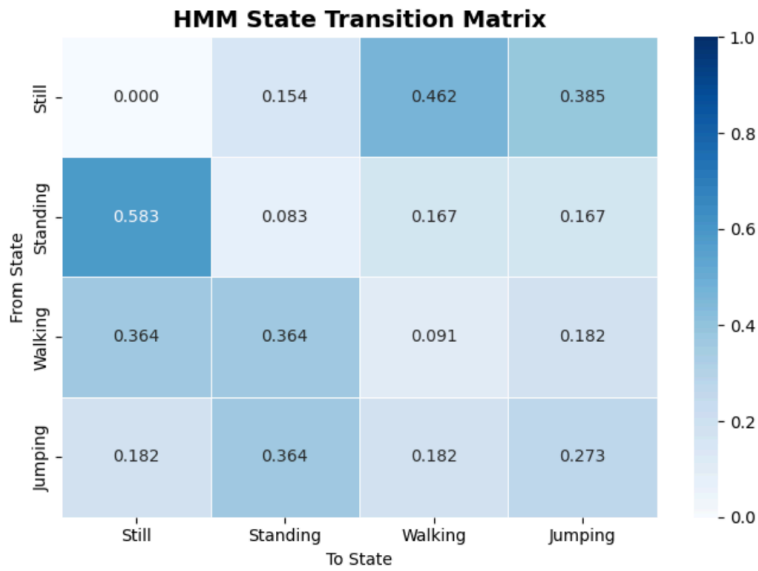
Parameter	Value
Number of Hidden States	4
Covariance Type	Diagonal
Maximum Iterations	50
Training Algorithm	Expectation-Maximization (Baum–Welch)
Decoding Algorithm	Viterbi

6. Visualization

Several visualizations were generated to understand model behavior.

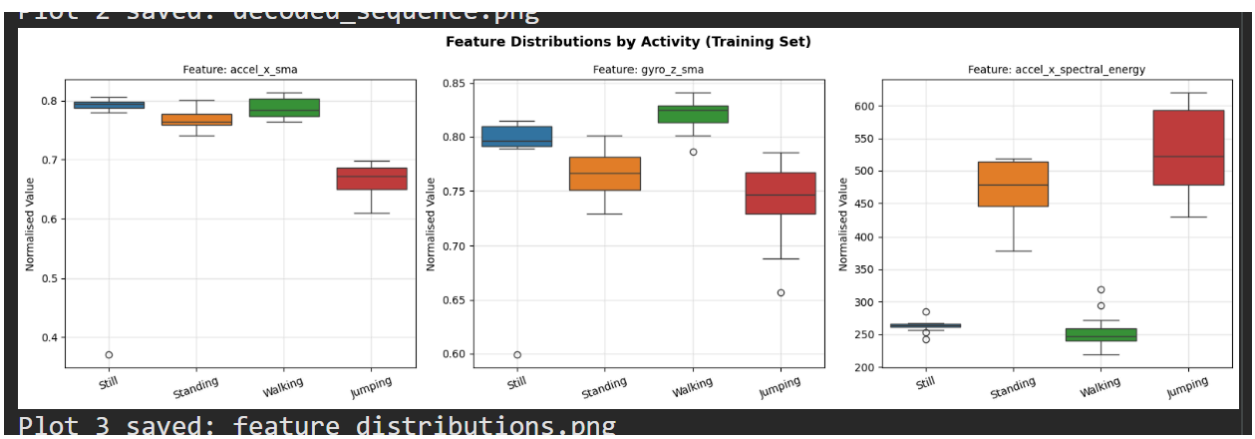
1. Transition Matrix Heatmap

Shows the probability of transitioning between activities.



2. Feature Distributions

Histograms of important features help illustrate how different activities produce distinct signal patterns.



7. Model Evaluation

The model was evaluated using both **window-level** and **clip-level** metrics.

Clip-Level Classification Report (Test Set)

Activity	Precision	Recall	F1-score	Support

Still	1.00	0.67	0.80	3
Standing	1.00	0.67	0.80	3
Walking	0.75	1.00	0.86	3
Jumping	0.75	1.00	0.86	3

8. Discussion

The HMM effectively captured the temporal structure of motion activities.

Strengths

- Models time-dependent behavior
- Handles noisy sensor data
- Works well with relatively small datasets

Challenges

- Similar dynamic movements can overlap in feature space
- Sensor noise and phone placement affect readings
- Jumping windows may resemble walking bursts

Potential Improvements

Several improvements could enhance the model:

- Collect longer activity recordings
- Increase dataset size
- Apply dimensionality reduction (PCA)
- Use Gaussian Mixture HMMs

9. Conclusion

This project demonstrates how Hidden Markov Models can be applied to recognize human activities from smartphone sensor data.By combining accelerometer and gyroscope signals with window-based feature extraction, the model successfully identified multiple daily activities.

Despite a relatively small dataset, the system achieved strong performance for most activities and effectively modeled temporal transitions between movements.This approach highlights the usefulness of probabilistic sequence models for real-world applications such as:

- health monitoring
- activity tracking

- context-aware mobile systems

Future work could improve robustness through larger datasets, additional sensors, and more advanced sequence models.

Github Repo: https://github.com/MikeManzi/hidden-markov-model

Dataset: markov

Team tasks: Team Task Sheet_ML Techniques II_Hidden Markov Models_Cohort 1_Team9]